College of Technology Galle

Course: Full Stack Software Development — Object Oriented Programming — Module K72T004M03 — NVQ Level 05

SEMESTER MODEL PAPER 09

Focus: Integrated OOP design under constraints, tracing, trade-offs

Difficulty: Hard

Duration: 3 Hours **Total Marks:** 100 **Questions:** 7 structured questions (answer any **5**) **Each question:** 20 marks

Instructions to Candidates

1. Answer any **five (5)** questions out of the seven provided.
2. Each question carries **20 marks**. Maximum mark for this paper is **100**.
3. Write answers in the answer booklet. Do **not** write answers on this question paper.
4. Where Java code is required, write clear, syntactically correct Java as taught in this module (VS Code / JDK).
5. Show workings / short explanations where asked. Manage your time (about 35 minutes per question).
6. Mobile phones and programmable devices are not allowed.

Question 1 [20 Marks]

Design under constraints.

a. Design a `Course` registration helper: course code, capacity, enrolled count. Enforce capacity with exceptions. **(8 marks)**

b. Implement enrol/drop methods with full validation. **(8 marks)**

c. Discuss whether capacity checks belong in the UI, the domain class, or both. **(4 marks)**

Question 2 [20 Marks]

Strategy via interfaces.

a. Create interface `DiscountPolicy` with `double apply(double price)`. **(4 marks)**

b. Implement `NoDiscount`, `PercentOff`, and `StudentDiscount`. **(9 marks)**

c. Write class `Cart` that holds a policy and a price, and show switching policies without changing Cart's core code. **(7 marks)**

Question 3 [20 Marks]

```
class Box {
    int n;
    Box(int n) { this.n = n; }
    void add(Box other) { this.n += other.n; }
}
// main:
Box x = new Box(5);
Box y = x;
Box z = new Box(5);
y.add(z);
System.out.println(x.n + " " + y.n + " " + z.n);
z = y;
z.n = 1;
System.out.println(x.n + " " + y.n + " " + z.n);
```

Harder trace.

a. Give both lines of output. **(8 marks)**

b. Explain aliasing after `Box y = x` and after `z = y`. **(8 marks)**

c. If `add` did `other = new Box(0)` inside the method, would `z` in `main` change? Why? **(4 marks)**

Question 4 [20 Marks]

Abstract factory-lite (manual).

a. Define abstract `Notification` with `deliver()` and concrete Email/SMS types. **(8 marks)**

b. Write a method `Notification create(String type)` that returns the right subclass (or throws on unknown type). **(8 marks)**

c. Explain how this supports open/closed thinking at beginner level. **(4 marks)**

Question 5 [20 Marks]

School system consistency.

a. Model Teacher, Classroom, and Student with a mix of association and composition - justify each link. **(8 marks)**

b. Implement enough Java to add a student to a classroom and list names. **(8 marks)**

c. What invariant should the model protect (e.g. student not in two seats)? How? **(4 marks)**

Question 6 [20 Marks]

Error-handling architecture.

a. Design a layered approach: domain throws, menu catches, logs a simple message. Illustrate with code sketches. **(10 marks)**

b. Should `main` catch `Exception` broadly? Argue for or against for this module's apps. **(6 marks)**

c. Name two unchecked and one checked exception category example (theory). **(4 marks)**

Question 7 [20 Marks]

```
class Bad {
    public int[] marks = new int[3];
    public String name;
    public double avg() {
        return (marks[0]+marks[1]+marks[2])/3.0;
    }
}
```

Refactoring challenge.

a. List four concrete problems in `Bad` (encapsulation, validation, API, misuse). **(8 marks)**

b. Produce an improved class API and implementation. **(8 marks)**

c. Show how client code must change and why that is healthier long-term. **(4 marks)**

******* END OF QUESTION PAPER *******